

Marketing Customer Analytics and Segmentation Project

Objective

The objective of this project was to analyze customer demographics, spending behavior, and marketing campaign performance to identify actionable business insights and customer segments.

Dataset

The dataset contains 2,240 customer records with demographic information, income levels, purchase behavior, and campaign responses.

Methodology

Data Cleaning

- Handled missing values in the Income column.
- Checked and removed data quality issues.
- Created new features including Age and TotalSpend.

Exploratory Data Analysis

- Analyzed customer age distribution.
- Examined income distribution.
- Evaluated spending patterns across education levels.
- Assessed marketing campaign performance.

Correlation Analysis

- Measured the relationship between Income and TotalSpend.
- Found a strong positive correlation of 0.66.

Customer Segmentation

- Applied K-Means Clustering.
- Used the Elbow Method to determine the optimal number of clusters.
- Detected and removed an extreme income outlier before final segmentation.

Key Findings

1. Most customers belong to the 40–60 age group.
2. Higher-income customers generally spend more.
3. Customers with PhD and Master's education levels demonstrate the highest spending behavior.
4. Campaign 4 was the most successful marketing campaign.
5. Three customer segments were identified:
 - Budget Customers
 - Regular Customers
 - VIP Customers

Recommendations

- Focus retention efforts on VIP customers.
- Design targeted promotional offers for Budget Customers.
- Upsell and cross-sell products to Regular Customers.
- Replicate strategies used in Campaign 4 for future marketing initiatives.

Technologies Used

Python, Pandas, NumPy, Matplotlib, Scikit-Learn, Jupyter Notebook





